Target validation paths — pancreatic-mets-kras-g12d-basal-arid1a-k9r3

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

This case turns on tissue. The molecular profile came from plasma, and two assays carry the most weight before any feature-targeting therapy is chosen: MMR protein IHC, which settles whether the immunotherapy door is open or closed, and a tissue NGS confirmation of the plasma KRAS G12D call, which the RAS(ON) inhibitor trials require to enroll. Two more assays harden the secondary axis and the trial paperwork: ARID1A/BAF250a IHC and high-resolution HLA class I typing. If MMR IHC shows retained protein, the dMMR/MSI-H checkpoint route stays closed and that low-yield option drops off the table without a wasted treatment course.

MLH1 / mismatch-repair status

Order MMR protein IHC for MLH1, PMS2, MSH2, and MSH6 on tumor tissue first. This is the gate on the whole immunotherapy question. The plasma call is a single somatic MLH1 variant in an MSS, TMB-8.6, germline-negative tumor, which usually means MMR is still proficient. Protein IHC settles it directly by showing retained versus lost nuclear staining, and retained expression argues against single-agent checkpoint blockade and closes the dMMR/MSI-H pembrolizumab and dostarlimab door. Read MLH1 and PMS2 together, since PMS2 is unstable without MLH1, and use the four-antibody panel rather than a PMS2/MSH6-only screen because the variant of interest sits in MLH1. IHC and MSI can disagree: a missense variant can leave a functional protein, and roughly 6% of MSI-H tumors retain MMR protein (pmid:32152268). Turnaround is 3 to 7 days once a block is in hand.

KRAS G12D

A tissue comprehensive genomic profile confirming KRAS G12D, with SMAD4, TP53, ARID1A, and MLH1 co-calls, is the second essential test. The driver came from plasma ctDNA, but the RAS(ON) inhibitor protocols want the mutation pathologically documented on tissue, so this confirmation hardens eligibility for daraxonrasib via NCT06625320, zoldonrasib, and MRTX1133, and rules out a ctDNA artifact before a trial slot is committed. The same report re-confirms the SMAD4, TP53, and ARID1A co-alterations that drive the rest of this workup, and the SMAD4 loss and TP53 status ride along for free as prognostic context. If no archival block exists, this row drives the decision about a liver-met core; weigh that against any biopsy reluctance with the treating team. Turnaround is 2 to 3 weeks from block receipt.

ARID1A

ARID1A/BAF250a IHC is high priority because the synthetic-lethal trials define ARID1A deficiency by loss of BAF250a protein, not by the DNA frameshift. In the ceralasertib study, only 67% (28/42) of ARID1A-mutant tumors showed BAF250a loss, so the frameshift call alone does not predict protein loss (pmid:41686845). This is the test that actually gates ATR-inhibitor entry on NCT03682289 and the EZH2 baskets such as NCT05023655. It runs off the same block as the MMR IHC with a 3 to 7 day turnaround. Many ARID1A baskets accept either a pathogenic mutation or BAF250a loss, but IHC loss is the stronger eligibility token where a protocol requires protein-level deficiency.

HLA-restricted KRAS immunotherapy

High-resolution HLA class I typing (A, B, C) by sequence-based typing documents the decision-relevant negative already in the file: the patient is HLA-C*05:01/C*06:02, not C*08:02, so the published C*08:02-restricted anti-KRAS-G12D TCR-T (NT-112 via NCT06218914; NCI NCT06690281) does not apply and should not be pursued. A clinical-grade, ASHI-accredited type satisfies trial enrollment paperwork and lets the screener match A *02:01- and A*01:01-restricted KRAS programs cleanly rather than relying on the research call alone. No tumor is needed: 5 to 10 mL of whole blood in EDTA or a buccal swab, with a 1 to 2 week turnaround.

SMAD4 and TP53 co-mutations

Two lower-priority blood-based steps round out the picture. Paired peripheral-blood-cell (buffy coat) sequencing adjudicates the plasma ASXL1 frameshift as clonal hematopoiesis versus tumor. ASXL1 is one of the top three CHIP genes, and a frameshift seen only in plasma cannot be assigned to tumor versus blood by VAF alone (pmid:32814631). If it is CHIP, drop it from the target list and the TMB accounting; this bundles with the whole-blood draw for HLA typing. The SMAD4 E520* loss and TP53 status come for free on the tissue NGS report already ordered for the KRAS confirmation, so order no separate assay for them. SMAD4 loss tracks with a metastasis-dominant, poorer-prognosis pattern and sharpens prognosis and DDR/cell-cycle trial stratification rather than naming a drug.

Where to order these assays

Assay	Provider	Decision gated	Contact
Mismatch-repair protein IHC (MLH1, PMS2, MSH2, MSH6) on tumor tissue	**Tempus (preferred) (MMR IHC reflex on xT/xF profiling)**	**Pembrolizumab/dostarlimab on the dMMR/MSI-H tumor-agnostic label; retained MMR expression argues against single-agent checkpoint blockade here.**	**test info · 600 W Chicago Ave, Suite 510, Chicago, IL 60654 · +1-800-739-4137**
Mismatch-repair protein IHC (MLH1, PMS2, MSH2, MSH6) on tumor tissue	Caris Life Sciences (MI Profile (MMR IHC))	Pembrolizumab/dostarlimab on the dMMR/MSI-H tumor-agnostic label; retained MMR expression argues against single-agent checkpoint blockade here.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · +1-888-979-8669
Mismatch-repair protein IHC (MLH1, PMS2, MSH2, MSH6) on tumor tissue	NeoGenomics Laboratories (MMR/MSI by IHC)	Pembrolizumab/dostarlimab on the dMMR/MSI-H tumor-agnostic label; retained MMR expression argues against single-agent checkpoint blockade here.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · +1-866-776-5907
Mismatch-repair protein IHC (MLH1, PMS2, MSH2, MSH6) on tumor tissue	Labcorp / Labcorp Oncology (MMR protein IHC panel)	Pembrolizumab/dostarlimab on the dMMR/MSI-H tumor-agnostic label; retained MMR expression argues against single-agent checkpoint blockade here.	test info · 358 South Main St, Burlington, NC 27215 · +1-800-845-6167
Tissue comprehensive genomic profiling (DNA NGS) confirming KRAS G12D, with SMAD4, TP53, ARID1A, MLH1 co-calls	**Foundation Medicine (preferred) (FoundationOne CDx)**	**RAS(ON) inhibitor trial entry (daraxonrasib/RMC-6236, RMC-9805, MRTX1133) requiring a pathologically documented KRAS G12D.**	**test info · 150 Second St, Cambridge, MA 02141 · +1-888-988-3639**

Tissue comprehensive genomic profiling (DNA NGS) confirming KRAS G12D, with SMAD4, TP53, ARID1A, MLH1 co-calls	Tempus (Tempus xT (tissue DNA/RNA))	RAS(ON) inhibitor trial entry (daraxonrasib/RMC-6236, RMC-9805, MRTX1133) requiring a pathologically documented KRAS G12D.	test info · 600 W Chicago Ave, Suite 510, Chicago, IL 60654 · +1-800-739-4137
Tissue comprehensive genomic profiling (DNA NGS) confirming KRAS G12D, with SMAD4, TP53, ARID1A, MLH1 co-calls	Caris Life Sciences (MI Cancer Seek / WES+WTS)	RAS(ON) inhibitor trial entry (daraxonrasib/RMC-6236, RMC-9805, MRTX1133) requiring a pathologically documented KRAS G12D.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · +1-888-979-8669
Tissue comprehensive genomic profiling (DNA NGS) confirming KRAS G12D, with SMAD4, TP53, ARID1A, MLH1 co-calls	MSK-IMPACT (Memorial Sloan Kettering) (MSK-IMPACT)	RAS(ON) inhibitor trial entry (daraxonrasib/RMC-6236, RMC-9805, MRTX1133) requiring a pathologically documented KRAS G12D.	test info · 1275 York Ave, New York, NY 10065 · +1-833-675-5437
ARID1A/BAF250a IHC (loss of nuclear expression)	**NeoGenomics Laboratories (preferred) (ARID1A (BAF250a) IHC)**	**ATR inhibitor (ceralasertib) and EZH2 inhibitor (tazemetostat) trials that require BAF250a loss by IHC.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · +1-866-776-5907**
ARID1A/BAF250a IHC (loss of nuclear expression)	Tempus (ARID1A IHC)	ATR inhibitor (ceralasertib) and EZH2 inhibitor (tazemetostat) trials that require BAF250a loss by IHC.	test info · 600 W Chicago Ave, Suite 510, Chicago, IL 60654 · +1-800-739-4137
ARID1A/BAF250a IHC (loss of nuclear expression)	Caris Life Sciences (ARID1A IHC)	ATR inhibitor (ceralasertib) and EZH2 inhibitor (tazemetostat) trials that require BAF250a loss by IHC.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · +1-888-979-8669
ARID1A/BAF250a IHC (loss of nuclear expression)	Labcorp / Labcorp Oncology (BAF250a (ARID1A) IHC)	ATR inhibitor (ceralasertib) and EZH2 inhibitor (tazemetostat) trials that require BAF250a loss by IHC.	test info · 358 South Main St, Burlington, NC 27215 · +1-800-845-6167
High-resolution HLA class I genotyping (A, B, C) by sequence-based typing	**Labcorp / Labcorp Transplant Diagnostics (preferred) (HLA class I high-resolution typing)**	**HLA-restricted KRAS G12D TCR-T / cell therapy (e.g. NT-112 via NCT06218914); confirms the patient is not C08:02 and so is ineligible for the C08:02-restricted products.**	**test info · 358 South Main St, Burlington, NC 27215 · +1-800-845-6167**
High-resolution HLA class I genotyping (A, B, C) by sequence-based typing	American Red Cross HLA Laboratory (High-resolution HLA class I typing)	HLA-restricted KRAS G12D TCR-T / cell therapy (e.g. NT-112 via NCT06218914); confirms the patient is not C08:02 and so is ineligible for the C08:02-restricted products.	test info · 100 Edgewood Ave NE, Atlanta, GA 30303 · +1-800-733-2767
High-resolution HLA class I genotyping (A, B, C) by sequence-based typing	Versiti Diagnostic Laboratories (HLA high-resolution typing)	HLA-restricted KRAS G12D TCR-T / cell therapy (e.g. NT-112 via NCT06218914); confirms the patient is not C08:02 and so is ineligible for the C08:02-restricted products.	test info · 638 N 18th St, Milwaukee, WI 53233 · +1-800-245-3117

High-resolution HLA class I genotyping (A, B, C) by sequence-based typing	NMDP / Histogenetics (reference HLA) (SBT high-resolution HLA)	HLA-restricted KRAS G12D TCR-T / cell therapy (e.g. NT-112 via NCT06218914); confirms the patient is not C08:02 and so is ineligible for the C08:02-restricted products.	test info · 300 Collins St, Ossining, NY 10562 · +1-914-682-4020
Paired peripheral-blood-cell (buffy coat) sequencing to adjudicate the plasma ASXL1 frameshift as CHIP vs tumor	**Tempus (preferred) (Paired tumor/normal (tissue + buffy coat))**	**Whether the plasma ASXL1 frameshift counts as a tumor alteration; CHIP origin removes it from target and biomarker accounting.**	**test info · 600 W Chicago Ave, Suite 510, Chicago, IL 60654 · +1-800-739-4137**
Paired peripheral-blood-cell (buffy coat) sequencing to adjudicate the plasma ASXL1 frameshift as CHIP vs tumor	Foundation Medicine (FoundationOne (germline/CHIP filtering))	Whether the plasma ASXL1 frameshift counts as a tumor alteration; CHIP origin removes it from target and biomarker accounting.	test info · 150 Second St, Cambridge, MA 02141 · +1-888-988-3639
Paired peripheral-blood-cell (buffy coat) sequencing to adjudicate the plasma ASXL1 frameshift as CHIP vs tumor	Guardant Health (Guardant360 (CH annotation))	Whether the plasma ASXL1 frameshift counts as a tumor alteration; CHIP origin removes it from target and biomarker accounting.	test info · 3100 Hanover St, Palo Alto, CA 94304 · +1-855-698-8887
Tissue confirmation of SMAD4 loss and TP53 status (read off the same tissue NGS report)	**Foundation Medicine (preferred) (FoundationOne CDx)**	**Prognostic weighting and DDR/cell-cycle trial stratification; no standalone drug gate.**	**test info · 150 Second St, Cambridge, MA 02141 · +1-888-988-3639**

Biomarker plan

Recommended assay	Rationale	Suggested assay provider	Tissue requirements

Mismatch-repair protein IHC (MLH1, PMS2, MSH2, MSH6) on tumor tissue	This is the gate on the whole immunotherapy question. The plasma call is a single somatic MLH1 variant in an MSS, TMB-8.6, germline-negative tumor, which usually means MMR is still proficient; protein IHC on tissue settles it directly by showing retained vs lost MLH1/PMS2 nuclear staining. Skip it and the team is guessing whether the dMMR/MSI-H pembrolizumab label even applies, when retained expression would close that door and spare a low-yield checkpoint course. Roughly 6% of MSI-H tumors retain MMR protein and a missense variant can leave a functional protein, so IHC and MSI can disagree (pmid:32152268).	Tempus (MMR IHC (reflex on xT/xF profiling)) · test info · 600 W Chicago Ave, Suite 510, Chicago, IL 60654 · +1-800-739-4137	archival FFPE block or unstained slides; no fresh biopsy needed
Tissue comprehensive genomic profiling (DNA NGS) confirming KRAS G12D, with SMAD4, TP53, ARID1A, MLH1 co-calls	The KRAS G12D driver came from plasma ctDNA, but the RAS(ON) inhibitor trials want the mutation pathologically documented on tissue, so a tissue NGS confirmation hardens eligibility and rules out a ctDNA artifact before committing to a trial slot. The RMC-9805 and daraxonrasib protocols specify a pathologically documented KRAS mutation, and tissue profiling simultaneously re-confirms the SMAD4, TP53, ARID1A, and MLH1 co-calls that drive the rest of this workup. Without it, a borderline-VAF plasma case risks a screen-fail or a wrong target call at the most decision-heavy step.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second St, Cambridge, MA 02141 · +1-888-988-3639	FFPE block or ~10-15 unstained slides; a liver-met core is acceptable if the primary block is unavailable

ARID1A/BAF250a IHC (loss of nuclear expression)	The synthetic-lethal trials in ARID1A-deficient tumors define deficiency by loss of BAF250a protein on IHC, not by the DNA frameshift alone, so this is the test that actually gates ATR- and EZH2-inhibitor enrollment. In the ceralasertib study only 67% (28/42) of ARID1A-mutant tumors showed BAF250a loss, so the frameshift call does not reliably predict protein loss (pmid:41686845). Order it and the ATR/EZH2 axis becomes a concrete trial option; skip it and the patient could screen-fail or be enrolled on an unconfirmed target.	NeoGenomics Laboratories (ARID1A (BAF250a) IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · +1-866-776-5907	archival FFPE block or unstained slides; same block as the MMR IHC can be used
High-resolution HLA class I genotyping (A, B, C) by sequence-based typing	The HLA genotype decides which KRAS G12D cell and TCR products can present the epitope, so a clinical-grade, high-resolution A/B/C type is the enrollment token for these programs. It confirms the decision-relevant negative already in the file: the patient is HLA-C05:01/C06:02, not C08:02, so the published C08:02-restricted anti-KRAS-G12D TCR-T (NT-112, NCT06218914; NCI NCT06690281) does not apply and should not be pursued. A confirmed type also lets the screener match A02:01- and A01:01-restricted KRAS programs cleanly instead of relying on the research HLA call alone.	Labcorp / Labcorp Transplant Diagnostics (HLA class I high-resolution typing) · test info · 358 South Main St, Burlington, NC 27215 · +1-800-845-6167	5-10 mL whole blood in EDTA, or a buccal swab; no tumor tissue needed

Paired (buffy coat) sequencing to adjudicate the plasma ASXL1 frameshift as CHIP vs tumor	ASXL1 is one of the top three CHIP genes, and a frameshift seen only in plasma cannot be assigned to tumor versus blood by VAF alone, so paired white-blood-cell sequencing is what separates clonal hematopoiesis from a true tumor variant (pmid:32814631). If the ASXL1 call is CHIP, it should be dropped as a tumor target and not counted toward TMB or trial stratification; if it is also present in tumor tissue, it stays in the co-alteration picture. The same paired analysis flags any other plasma calls that are really hematopoietic in origin.	Tempus (Paired tumor/normal (tissue + buffy coat)) · test info · 600 W Chicago Ave, Suite 510, Chicago, IL 60654 · +1-800-739-4137	5-10 mL whole blood for buffy-coat DNA; pairs with the tissue NGS draw
Tissue confirmation of SMAD4 loss and TP53 status (read off the same tissue NGS report)	SMAD4 loss tracks with a metastasis-dominant, poorer-prognosis pattern and TP53 inactivation is near-universal in PDAC, so confirming both on tissue mainly sharpens prognosis and DDR/cell-cycle trial stratification rather than naming a drug. These calls come for free on the tissue NGS panel already ordered for KRAS confirmation, so no separate specimen or cost is involved. The value is making sure the prognostic weighting and any combination rationale rest on tissue rather than plasma alone.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second St, Cambridge, MA 02141 · +1-888-988-3639	none beyond the tissue NGS block already submitted

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.